

Worksheet: 09 Switch Factory Default Reset

Leeruitkomsten (wat moet je kennen / kunnen):

- Je kan een CISCO switch resetten naar de Factory Defaults

Wat moet je kennen:

CISCO IOS

Een switch is een apparaat dat netwerkverkeer beheert tussen apparaten in een LAN (Local Area Network). Het werkt op Layer 2 (datalink layer) en soms op Layer 3 (network layer) van het OSI-model. Een reset is noodzakelijk in de volgende situaties:

- **Wachtwoord vergeten:** Toegang tot de switch is geblokkeerd
- **Herconfiguratie:** De switch moet opnieuw worden ingericht
- **Troubleshooting:** Problemen oplossen door opnieuw te beginnen met een schone configuratie

Een CISCO switch kan voorzien zijn van **CISCO IOS en IOS-XE:**

- **IOS:** Traditionele software op Cisco-switches en routers, draait direct op de hardware
- **IOS-XE:** Moderne versie van IOS, gebouwd op een Linux-kern, biedt betere schaalbaarheid en functies zoals packages en containers

IOS commando's:

IOS reset procedures in ROMMON mode

Task	IOS Command
Reset Catalyst IOS Switches to Factory Defaults in ROMMON mode	
Power off the switch and hold down the Mode button. While holding down the Mode button power on the switch. Hold down the Mode button until you see the output Switch: NOTE: some switches displays the message "The password-recovery mechanism is enabled". Press MODE button twice before you see the output Switch:	Boot into ROMMON mode Switch:
Loads flash memory	Switch: flash_init
Removes config.text	Switch: delete flash:config.text
Removes vlan.dat	Switch: delete flash:vlan.dat
The vlan.dat file is a configuration file used by Cisco devices to store VLAN-related information. It is a binary file stored in the flash memory of a Cisco switch <i>!-- Make sure vlan.dat and config.text both are deleted --!</i>	
Reboot switch	
Reboots the switch	Switch: reset
or	
	Disconnect and reconnect the power cord
Initial Configuration Dialog (if it is shown)	
Device is empty when showing the Initial Configuration Dialog (if it is shown): "Would you like to enter the initial configuration dialog [yes/no]" Always enter: no	

IOS-XE reset procedures in ROMMON mode

Task	IOS Command
Reset Catalyst IOS-XE Switches to Factory Defaults in ROMMON mode	
Power off the switch and hold down the Mode button . While holding down the Mode button power on the switch. Hold down the Mode button until wait for SYST and ACTV LED to become amber before releasing MODE button	Boot into ROMMON mode Switch:
Loads flash memory	Switch: flash_init
Bypass STARTUP-CONFIG in NVRAM	Switch: SWITCH_IGNORE_STARTUP_CFG=1
Reloads the switch	Switch: boot
Initial Configuration Dialog (if it is shown)	
Device is empty when showing the Initial Configuration Dialog (if it is shown): "Would you like to enter the initial configuration dialog [yes/no]"	
Always enter: no	
Erase config.text and VLAN.dat in privilege exec mode	
Shows content of NVRAM	Switch# dir flash:
Removes vlan.dat (if exists)	Switch# delete flash:vlan.dat
The vlan.dat file is a configuration file used by Cisco devices to store VLAN-related information. It is a binary file stored in the flash memory of a Cisco switch	
!-- Make sure vlan.dat and config.text both are deleted --!	
Save empty configuration	
Copy the new and empty running-config to the old startup-config	Switch# copy running-config startup-config or Switch# write
Changing IGNORE_STARTUP setting back	
Sets IGNORE_STARTUP back to original value	Switch# no system ignore startupconfig switch all
Reboots the switch	Switch# reload or Disconnect and reconnect the power cord
or	
Power off the switch and hold down the Mode button . While holding down the Mode button power on the switch. Hold down the Mode button until wait for SYST and ACTV LED to become amber before releasing MODE button	Boot into ROMMON mode Switch:
Instructs the switch to read/load the startup-config	Switch: SWITCH_IGNORE_STARTUP_CFG=0
Reboots the switch	Switch: boot or Disconnect and reconnect the power cord
Device is empty when showing the Initial Configuration Dialog (if it is shown): "Would you like to enter the initial configuration dialog [yes/no]"	
Always enter: no	

IOS and IOS-XE reset procedures in Privileged EXEC Mode

Task	IOS Command
Erase config.text and VLAN.dat in privilege exec mode	
Shows content of NVRAM	Switch# dir flash:
Erases config.text	Switch# erase startup-config
	or
	Switch# write erase
Removes vlan.dat	or
	Switch# delete flash:config.text
	Switch# delete flash:vlan.dat
The vlan.dat file is a configuration file used by Cisco devices to store VLAN-related information. It is a binary file stored in the flash memory of a Cisco switch	
!-- Make sure vlan.dat and config.text both are deleted --!	
Reboots the switch	Switch# reload
	or
	Disconnect and reconnect the power cord
Initial Configuration Dialog (if it is shown)	
Device is empty when showing the Initial Configuration Dialog (if it is shown): "Would you like to enter the initial configuration dialog [yes/no]"	
Always enter: no	

Verifying configuration Commands

Task	IOS Command
Verify Basic Configuration	
Shows the current configuration file stored in RAM	Switch# show running-config
Shows the saved configuration stored in config.text located in NVRAM	Switch# show startup-config
Displays information on all configured VLANs	Switch# show vlan brief

Default VLANs by factory reset are:

Switch# **show vlan brief**

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gi0/1, Gi0/2
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

Wat moet je kunnen:

Je kan een CISCO switch resetten.

Doel:

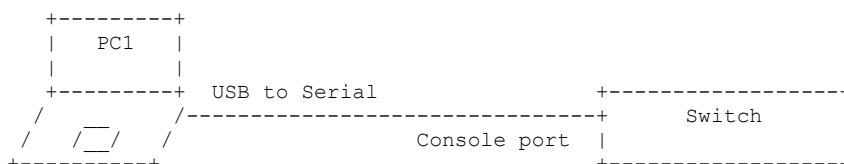
Je kan een CISCO switch met IOS via de ROMMON mode.

Praktijkvoorbeeld:

Stel dat een organisatie oude switches heeft ontvangen die volledig opnieuw geconfigureerd moeten worden. Je moet de switch resetten naar fabrieksinstellingen. Met de volgende vereisten:

- De computer is, met een consolekabel, verbonden met de Console poort van de switch
- Op het systeem is TERA TERM of PUTTY geïnstalleerd

Topologie:

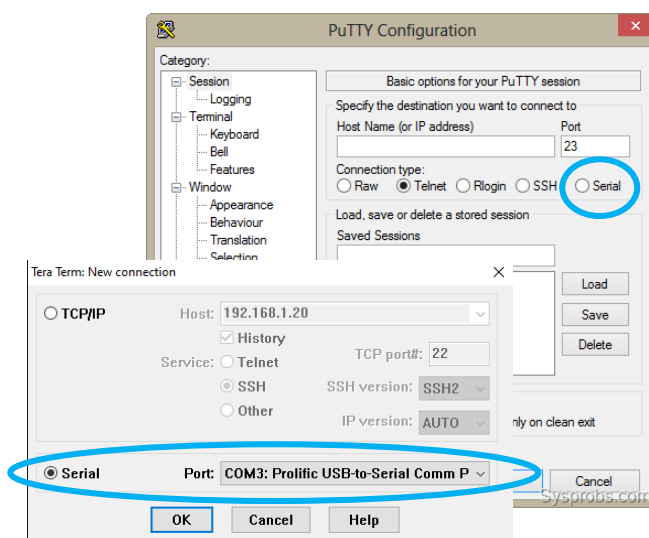


Configuratie:

Stap 1: Verbind de laptop met de switch via de console-kabel

Stap 2: Start Tera Term of PuTTY

- Kies **Serial Connection** en stel de juiste COM-poort in
- Configureer eventueel de baudrate op **9600**
- Log in via de console en maak connectie



Stap 3: Start op in de ROMMON Mode

Schakel de switch uit en houd de **Mode-knop** ingedrukt. Terwijl je de **Mode-knop** ingedrukt houdt, schakel je de switch weer in

Houd de Mode-knop ingedrukt totdat het system LED amber wordt of je de uitvoer **Switch:** ziet. Je zit nu in de ROMMON Mode



Stap 4: Laad het flash geheugen

Switch: **flash_init**

Stap 5: Verwijder config.text en vlan.dat

Switch: **dir flash:**

Switch: **delete flash:config.text**

Switch: **delete flash:vlan.dat**

Stap 6: Herstart de switch

Switch: **reset**

Stap 7: Onderbreek de Initial Configuration Dialog

Als de switch opgestart is en succesvol teruggezet is naar de Factory Defaults zie je de volgende melding:

"Would you like to enter the initial configuration dialog [yes/no]"

Always enter:

No

Je komt nu in de User EXEC Mode en kan de switch configureren

Switch>